

# G6PD Deficiency

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## G6PD Deficiency in Children

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### Outline

Introduction & inheritance · What G6PD does · Why older RBCs are hit first · What destroys the RBC · Variants · Favism · Clinical features · Risk factors (triggers) · Balanced polymorphism & malaria · Investigations · Treatment & management

### Introduction

**G6PD (glucose-6-phosphate dehydrogenase)** is an **enzyme inside red blood cells** whose job is to **protect RBCs from damage**. If G6PD is missing or low, **RBCs become weak against stress and break (haemolysis)**. It is therefore the **most common RBC enzyme defect** and causes a **haemolytic anaemia**.

**Inheritance: X-linked recessive** – so **males (XY)** are more affected, while **females (XX)** are usually carriers. It is common in **Africans, Asians and Mediterranean populations** – because of **malaria protection** (see below).

## What exactly does G6PD do?

G6PD's main job is to **help produce NADPH**. NADPH keeps **glutathione in its reduced form**, which **protects RBCs from oxidative stress** (reactive oxygen species, ROS).

**The chain:** G6PD → NADPH → reduced glutathione → protects the RBC.

If G6PD is deficient: **down NADPH → down reduced glutathione → up oxidative damage**. The result is that **haemoglobin is damaged, the RBC membrane is damaged, and the RBC breaks – haemolysis**.

## Why are older RBCs worse affected?

Older RBCs already have **less enzyme activity** – they've used it up, and because mature RBCs have **lost their nucleus**, they **cannot synthesise new protein (new G6PD)**. So each mature red cell is left with a **limited, dwindling amount of G6PD**, and as it ages the enzyme is used up. Older RBCs therefore **cannot handle stress at all** and are **destroyed first** in acute haemolysis.

## What actually destroys the RBC?

**Oxidative stress** (very important). Sources: **drugs (e.g. antimalarials), infections, and chemicals (like fava beans)**.

Inside the RBC: (1) oxidants accumulate → (2) **haemoglobin is damaged, forming Heinz bodies** → (3) the RBC becomes abnormal → (4) the **spleen removes it by biting off the Heinz-body part** (forming “bite cells”, seen on the blood smear) or the cell bursts. The result is **haemolysis and anaemia**.

## Variants of G6PD (simple understanding)

- **G6PD B+** – the normal variant in most populations.
- **G6PD A+** – common in African-Americans; there is **reduced activity but the enzyme is partially functional**.
- **Severe variants** – markedly reduced G6PD activity, more dangerous.

## Special condition: favism

Eating **fava beans** triggers **massive oxidative stress**. Fava beans contain **vicine and convicine**, which are **hydrolysed in the gut to divicine**, a **potent oxidant**. The result is **sudden severe haemolysis**.

## Clinical features

Most individuals are **asymptomatic**. Symptoms arise **48-96 hours (2-4 days)** after **ingestion of the offending drug** – because the reactive oxygen species must **accumulate to a high enough concentration** to trigger the crisis. **Severity depends on the level of enzyme defect**.

**Symptoms:** pallor (anaemia), jaundice (bilirubin from RBC breakdown), fatigue, and sometimes dark urine.

**Key idea:** the patient is **NORMAL between episodes**.

## Risk factors (triggers)

1. **Infections (most common)** – the body's phagocytes (neutrophils, macrophages) produce a lot of ROS to fight infection. In a normal child, RBC glutathione neutralises this; in a G6PD-deficient child, it causes **acute haemolysis**.
2. **Oxidant drugs** – e.g. antimalarials (primaquine), sulfa drugs.
3. **Fava beans**.

## Balanced polymorphism & malaria (high-yield)

**Balanced polymorphism** is the maintenance of two or more alleles in a population because the **heterozygote has a survival advantage**. In mild G6PD deficiency, RBCs are fragile – but the **malaria parasite cannot survive well**, because any little oxidative stress (from the plasmodium itself) makes the fragile G6PD-deficient RBC **haemolyse**, destroying the infected cell early. This gives **partial protection against malaria**, explaining why G6PD deficiency is common in malaria-endemic areas.

## Investigations

Diagnosis is confirmed by **evidence of reduced G6PD activity in RBCs** (enzyme activity  $\leq 10\%$  of normal confirms it), measured by **spectrophotometry** or newer techniques (RDTs).

**Important:** the test is **best done when the patient is stable, not during a crisis**. During haemolysis the old (deficient) cells are destroyed and new reticulocytes have **normal enzyme**, which can give a **falsely normal result**. Best performed when the **reticulocyte count is normal**.

## Treatment & management

The condition is **self-limiting** – once the trigger is removed, haemolysis stops. **Blood transfusion** may be required in severe cases.

**Management:** (1) **avoid triggers**, (2) **treat infection**, and (3) in **severe cases, blood transfusion**.

## References

Lecture material of Prof I.F. Ogbonna (500-level Paediatrics, Abia State University); Nelson Textbook of Paediatrics.